Analyzing Crime Patterns and Forecasting Trends

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Introduction

Crime forecasting has become an essential tool for keeping people safer while also assisting police in making decisions. Crime trends in cities are typically affected by time as well as location, thus it is important to know how crime changes over time. Being able to accurately anticipate crime patterns may help police use their resources more effectively, identify areas more likely to experience crime, and develop proactive measures to prevent crime before it happens.

The main goal of this research is to look at crime statistics from Los Angeles from 2020 to 2024. The main purpose is to identify patterns in crime and predict future events. The dataset contains precise records of crimes, including the date, time, and type of crime. This makes it possible to do both temporal and statistical analysis.

A significant issue examined in this work is the difficulty of modeling crime data that displays both long-term trends and short-term variations. In general, crime levels may follow trends that are easy to forecast, but daily or monthly changes are typically random and influenced by external factors such as socioeconomic conditions, the time of year, or unforeseen events. This makes it hard to generate reliable predictions and requires models to handle both structure and uncertainty.

To address this problem, this project applies two different modeling approaches:

- Time Series Modeling (AR model) to capture patterns and trends in crime statistics over time
- Bayesian modeling to account for uncertainty and use probabilistic approaches to figure out crime distributions

By comparing these approaches, the project aims to evaluate how well each method predicts crime trends and to understand their strengths and limitations in handling real-world data.

Literature Review

Recent research has shown an increasing interest in using machine learning and statistical techniques for crime prediction. Numerous research projects focus on using historical crime data to identify trends and predict future incidents, aiming to enhance public safety and optimize resource allocation.

Alsubayhin et al. (2024) examined the use of classification approaches, including Random Forest, Logistic Regression, and LightGBM, to forecast crime categories and probabilities. Their results indicate that machine learning models may effectively capture intricate relationships in crime data and improve predictive accuracy compared with conventional methodologies. This shows how important it is to use sophisticated models when working with big, complicated datasets.

The article (*Crime Prediction Using Machine Learning and Deep Learning: A Systematic Review and Future Directions*, 2023) is a more comprehensive analysis of over 150 studies that identified prevalent methodologies, including linear regression, support vector machines (SVMs), and decision trees, highlighting the importance of data quality and feature selection in shaping model performance. Another review, Jenga et al. (2023), also shows that most models rely heavily on spatial and temporal features, such as time and place, to improve prediction accuracy.

The project contributes to the body of knowledge by using Bayesian regression and time-series modeling to predict crime rates in Los Angeles. This study focuses on interpretability and comparing approaches, which are different from much of the other research that concentrates on classification or deep learning. The study offers insights into how various modeling strategies address temporal patterns and uncertainty in crime data by evaluating the strengths and weaknesses of each.

Machine Learning Methodology and Data Collection

<u>Data Collection and Preparation</u>:

This project used a dataset from the City of Los Angeles open data portal (data.gov) with detailed records of crimes that occurred between 2020 and 2024. The date and time of the crime, the place where it happened, and the type of crime are all included in each record. We chose this dataset because it has both time and space features, which are important for studying crime patterns and making predictions about the future.

The data were combined into daily crime counts to help with the problem of predicting crime trends. This change makes the dataset easier to read by converting it to a time-series format, where each observation represents the total number of crimes per day.

The data were then split into:

- Training data (2020 - 2023) for model development
- Testing data (2024) for evaluating prediction performance

This setup allows us to compare predicted values with actual observations and assess how well each model generalizes to future data.

<u>Methodology Overview:</u>

This project applies two main modeling approaches, Time Series Modeling and Bayesian Modeling. These methods were chosen to compare a traditional statistical approach with a probabilistic approach that accounts for uncertainty.

Machine Learning Methodology and Data Collection

1. Time Series Modeling (AR Model):

The time series model used in this project is an Autoregressive model of order p (AR(p)), which predicts future values based on past observations.

Dataset: Daily crime counts in Los Angeles from 2020 to 2024

Preprocessing: This was first processed by converting dates to datetime format, sorting chronologically, and aggregating incidents into daily totals.

Analysis: Explored the temporal structure of the data using Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to determine the appropriate number of lag values.

Model: On this analysis, we selected 14 lags, resulting in an AR(14) model. Lag features (1-14) were manually constructed, and the model parameters were estimated using least squares regression.

Prediction: Predictions for 2024 were generated using a recursive forecasting approach, in which each predicted value is used as input to subsequent predictions.

Evaluation: The dataset was split into training data (2020-2023) and testing data (2024). Model performance was assessed by comparing predicted crime counts with actual observed values.

2. Bayesian Model:

The Bayesian model is used to predict the number of crimes for the whole of Los Angeles and for each area.

Preprocessing: Sorting the data by time, adding a new column for a cleaner date, and aggregating by date and area. Using data from 2020 to 2023 as a training set and 2024 data as a testing set.

Predicting the crime counts of the city of Los Angeles in 2024:

- We are establishing the prior distribution using the Poisson distribution, incorporating features such as DayOfWeek, Month, IsWeekend, and WeekOfYear.
- The Poisson distribution is an appropriate choice because it models the probability of a specific number of events (in this case, crimes) occurring within a fixed time interval (the year 2024). Daily crime counts are discrete and non-negative integers with no predetermined upper bound. The Poisson assumption that events occur independently at a constant average rate is reasonable for daily crime aggregates across a large metropolitan area, where individual crimes are independent events driven by the overall crime rate.

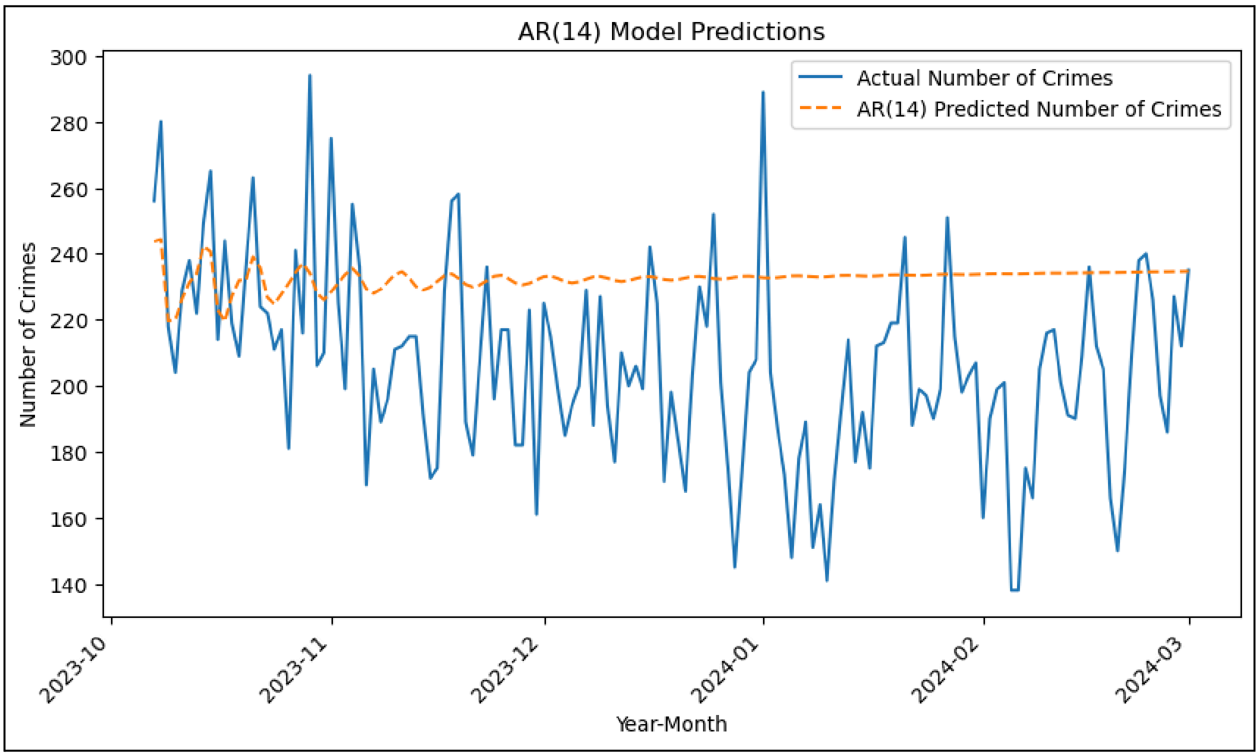
Predicting the crime counts for a chosen area:

- We are establishing the prior distribution using the Poisson distribution, incorporating DayOfWeek, MonthNum, and AreaName.
- The Poisson distribution is applied to model crime counts within specific areas of Los Angeles, as each area experiences crime incidents as discrete events that accumulate over a given observation period (one day).

Machine Learning Results and Interpretation

Time Series Model:

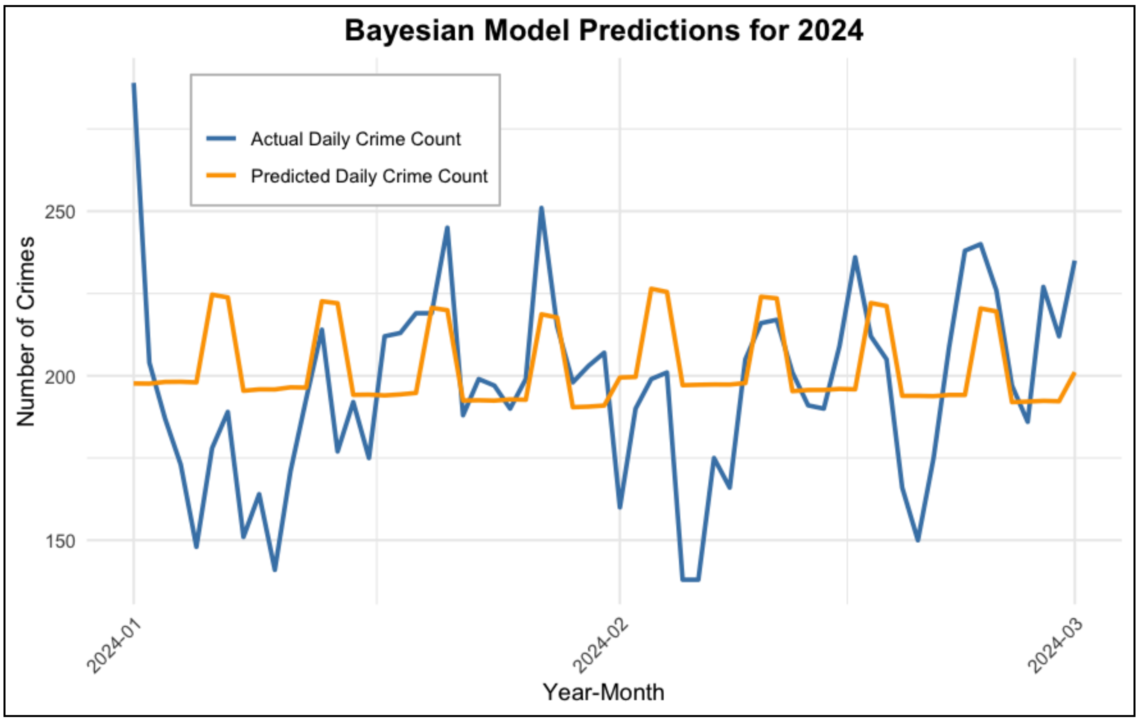
- The AR(14) model captures the overall average trend in crime counts but produces overly smooth predictions. However, it fails to capture sharp fluctuations, indicating limited performance for highly variable crime data.
- The model captures the time-series structure, but the flat prediction profile indicates that ARIMA struggles with the high day-to-day variability inherent in crime data.



Bayesian Model:

Overall crime counts in Los Angeles:

- The Bayesian model yielded a result quite similar to that of the Time Series model. The predicted values closely tracked the general trend of actual crime counts, maintaining estimates in the 190-225 range that reflected the underlying baseline activity.
- However, the Bayesian model also had difficulty capturing day-to-day fluctuations, as did the Time Series model.



Crime counts for a specific area in Los Angeles:

- Using Hollywood in February 2024 as a test case, the model produced a posterior predictive mean of 12.38, compared to the observed mean of 12.03. This indicates strong predictive accuracy for that month.

```
> # See the mean of predicted number of crime
> mean(predictions_specific$Predicted)
[1] 12.3821
> # See the mean of true number of crime
> mean(predictions_specific$Count)
[1] 12.03448
```

- Model performance varied substantially across months, with February achieving notably lower prediction error (MAE = 3.18) compared to other months (MAE = 11-12).
- This difference suggests that February crime patterns are more stable and predictable, likely due to fewer major events, more consistent weather, and reduced holiday-related disruptions that characterize other months.

```
> # Calculate the mean absolute error
> mae <- mean(abs(predictions_specific$Count - predictions_specific$Predicted))
> print(paste("MAE:", round(mae, 2)))
[1] "MAE: 3.18"
```

Comparing between 2 models:

- Overall, both models can predict the general trend in crime counts for 2024. However, both faced the same difficulty of capturing the day-to-day fluctuations.
- For the Time Series model, we should explore multivariate time series to model interactions among crime types across areas.
- For the Bayesian model, we consider adding more socioeconomic factors such as the unemployment rates, poverty levels, housing instability, etc., to better understand the structural issues that drive crime.

Conclusions and Future Work

Conclusions:

This study analyzed crime data from Los Angeles (2020–2024) employing time series and Bayesian modeling to identify patterns and predict trends. Both models were able to show the general trend in the number of crimes. The AR(14) model used past values to make predictions, while the Bayesian model provided good average estimates and accounted for uncertainty. However, both approaches struggled to capture short-term changes. The AR model made smooth predictions but missed sudden changes, while the Bayesian model usually predicted values close to the mean. This shows that crime patterns are affected by more than just time.

The assumptions of each model were only partially fulfilled. The AR model assumes stationarity and linear correlation, which may not hold for real-world crime data. The Bayesian Poisson model presupposes a constant rate, along with consistent mean and variance, which may not accurately reflect the variability in crime statistics. Even with these problems, both models were good starting points for figuring out general trends.

Future Work:

Future work could improve this analysis by incorporating additional variables such as socioeconomic factors, seasonal effects, or location-based features. More advanced models, such as ARIMA or machine learning methods, could better capture complex patterns and variability. Using models that address problems such as non-stationarity and overdispersion (e.g., Negative Binomial models) could also help. Including spatial patterns from different areas in the analysis would also give more detailed information.

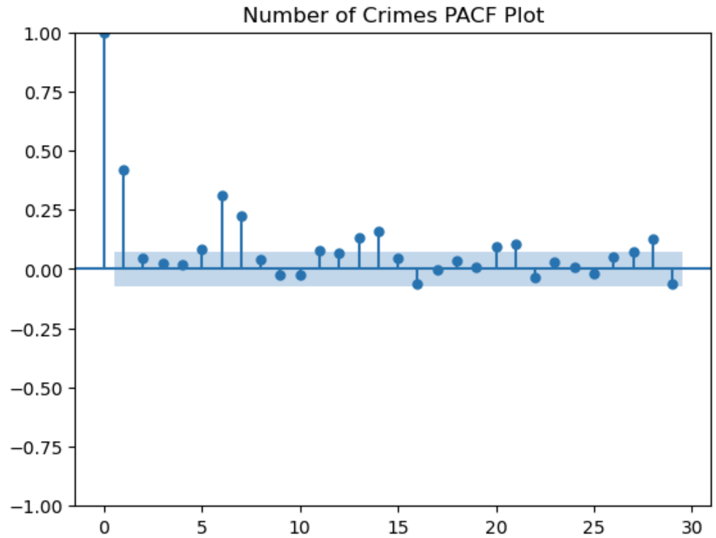
Overall, while this study provides a useful foundation, further improvements are needed to better model the complexity of crime data and produce more accurate predictions.

Appendix: Implementation

Time Series:

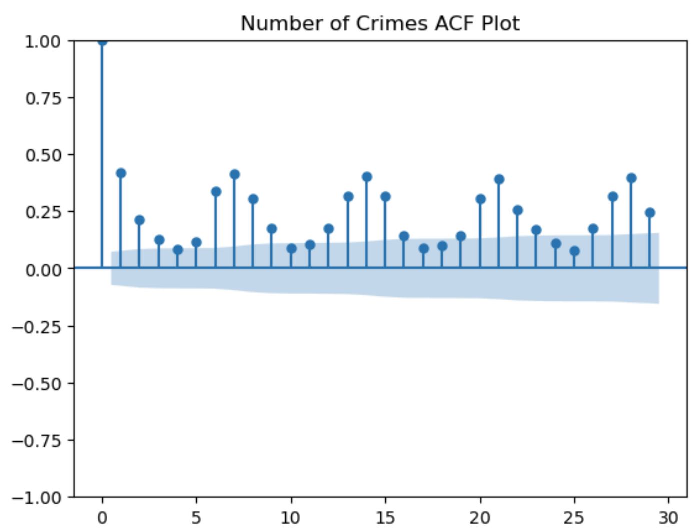
In [57]:

```
# Get the PACF plot
plot_pacf(series, title="Number of Crimes PACF Plot")
plt.show()
```



In [56]:

```
# Get the ACF Plot
series = crimes_2022_2024["Counts"]
plot_acf(series, title="Number of Crimes ACF Plot")
plt.show()
```



```
In [58]: # Prepare the data, split into 80% training set and 20% test set
         train_size = int(0.8 * len(crimes_2022_2024))
         train_data = crimes_2022_2024[:train_size]
         test_data = crimes_2022_2024[train_size:]

         y_train = np.array(train_data["Counts"]).reshape(-1, 1)
         y_test = np.array(test_data["Counts"]).reshape(-1, 1)
```

```
In [60]: # Training values
         p = 14
         train_ar = train_data.copy()

         for lag in range(1, p + 1):
             train_ar[f"Lag_{lag}"] = train_ar["Counts"].shift(lag)

         train_ar = train_ar.dropna()
         lag_cols = [f"Lag_{lag}" for lag in range(1, p + 1)]
         X_train = train_ar[lag_cols].values
         y_train = train_ar["Counts"].values
         beta_hat = np.linalg.lstsq(X_train, y_train, rcond=None)[0]
         history = list(train_data["Counts"].values)

In [61]: # Predictions
         y_pred = []
         for t in range(len(test_data)):
             x_input = np.array(history[-p:][::-1])
             yhat = np.dot(beta_hat, x_input)
             y_pred.append(yhat)
             history.append(yhat)

In [62]: # Put predictions into a data frame
         pred_data = pd.DataFrame({
             "Counts": y_pred
         }, index=test_data.index)
```

Bayesian Model:

```
# Split: Train = 2020-2023, Test = 2024
train_data <- crimes %>% filter(Year < 2024)
test_data <- crimes %>% filter(Year == 2024)

# Define priors
crime_priors <- c(
  prior(normal(5.5, 0.5), class = "Intercept"),
  prior(normal(0, 0.3), class = "b")
)
```

```r
# We use Poisson distribution here because the number of crimes
# happened every day is independent from each other
# and are counted in non-negative integers
model <- brm(
  Count ~ DayOfWeek + Month + IsWeekend + WeekOfYear,
  family = poisson(),
  data = train_data,
  prior = crime_priors,
  chains = 4,
  iter = 2000,
  warmup = 180
)

# Check model summary
summary(model)

plot(model)

# Point predictions (posterior mean)
predictions_mean <- predict(model, newdata = test_data)[, "Estimate"]

# Full posterior predictive distribution
predictions_full <- posterior_predict(model, newdata = test_data)

# Get credible intervals
pred_intervals <- predictive_interval(model, newdata = test_data, prob = 0.95)
```

```r
# Define priors
area_prior <- c(
  prior(normal(5, 1), class = "Intercept"),
  prior(normal(0, 5), class = "b")
)

# We use negative poisson models
model_area <- brm(
  Count ~ DayOfWeek + MonthNum + AREA.NAME,
  family = poisson(),
  data = train,
  prior = area_prior,
  chains = 4,
  iter = 1000,
  warmup = 180
)
summary(model_area)
plot(model_area)
```